KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 21 · SCIENCE

Physical and Chemical Changes: Where Does the Matter Go?

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: What Do I Already Know?

Answer with what you know right now. It is OK to guess!

1. **Review (Week 19):** Undisturbed rock layers, top to bottom, are sandstone with tracks, limestone with corals and shale with fern prints. Which layer formed first? What environment did the limestone form in?

2. What do you think is the difference between a physical change and a chemical change? Give one example of each.

3. Predict: A candle burns on a scale for an hour. Will the scale show more, less or the same mass? Where do you think the missing wax goes?

4. Circle the changes you think make a NEW substance: ice melting · toast turning brown · a nail rusting · sugar dissolving · paper burning

5. Draw what you think happens to the tiny particles of a substance during a chemical change. Label your drawing.

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POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Account for Every Gram

Use evidence from today's lesson. Write in complete sentences.

1. Bianca sees bubbles in two cups: water boiling on the stove, and cold vinegar with baking soda. Which cup shows a chemical change? Explain using the words **new substance**.

2. **Two-step problem.** An open dish of reactants weighs 86.4 grams. After the reaction it weighs 82.9 grams. How many grams of gas escaped? What would a sealed jar show?

3. **Find the mistake.** Boaz says: "Burning steel wool gets heavier, so it creates new matter." Explain.

4. Why did the Statue of Liberty's copper stay strong while an iron gate crumbled? Use the word **patina**.

★ **Challenge:** Draw 8 toy bricks as two reactants, then the same bricks rebuilt as one product. Explain how this models conservation of mass.

Looking Ahead: Next week: Senses and Signals. How do animals sense the world, send signals through nerves to the brain and respond?